Honors Calculus 1610

Lecture 13: Limits and Derivatives of Vector-Valued Functions

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Today

- One more limit of a vector-valued function, done by hand.
- Limits are computed componentwise — and why.
- The derivative of a vector-valued function, and the approximation $\vec{f}(c+h) \approx \vec{f}(c) + h\,\vec{f}\,'(c)$.
- The rules: sums, scalar multiples, and two Leibniz rules.
- For a curve on a sphere, $\vec{f}\,'(t)$ is perpendicular to $\vec{f}(t)$.

Last time

We considered functions $\vec{f}\colon (a,b) \to \mathbb{R}^n$.

Definition

Let $c \in (a,b)$. We say $\lim_{t\to c} \vec{f}(t) = \vec{L}$ to mean: for all $\epsilon > 0$ there exists $\delta > 0$ such that

$$0 < |t-c| < \delta \quad \Longrightarrow \quad \|\vec{f}(t) - \vec{L}\| < \epsilon.$$

Exactly the old definition, with the absolute value on the right replaced by the length.

Example

$$\vec{f}(t) = \left(\frac{t^2 - 1}{t - 1},\ 2t + 1 \right), \qquad \text{claim: } \lim_{t \to 1} \vec{f}(t) = (2, 3) = \vec{L}.$$

Scratch work. For $t \neq 1$ we have $\frac{t^2-1}{t-1} = t + 1$, so

$$\|\vec{f}(t) - (2,3)\| = \|(t-1,\ 2t-2)\| = |t-1|\,\|(1,2)\| = \sqrt{5}\,|t-1|,$$

and this is $< \epsilon$ exactly when $|t - 1| < \dfrac{\epsilon}{\sqrt{5}} = \dfrac{\sqrt{5}}{5}\,\epsilon$.

The middle step pulls the scalar out of the length: $\|\lambda \vec{a}\| = |\lambda|\,\|\vec{a}\|$.

Example: the proof

Choose $\epsilon > 0$, and let

$$\delta = \frac{\sqrt{5}}{5}\,\epsilon.$$

If $0 < |t-1| < \delta$, then $t \neq 1$, and

$$\sqrt{5}\,|t-1| < \sqrt{5} \cdot \frac{\sqrt{5}}{5}\,\epsilon = \epsilon.$$

Copying the scratch work backwards,

$$\|\vec{f}(t) - (2,3)\| = \sqrt{5}\,|t-1| < \epsilon. \qquad \square$$

Limits are componentwise

Proposition

Let $\vec{f}\colon (a,b) \to \mathbb{R}^n$ be a vector-valued function, and write

$$\vec{f}(t) = (f_1(t), f_2(t), \ldots, f_n(t)), \qquad f_i\colon (a,b) \to \mathbb{R}.$$

Then $\lim_{t\to c} \vec{f}(t) = \vec{L} = (L_1, L_2, \ldots, L_n)$ if and only if $\lim_{t\to c} f_i(t) = L_i$ for every i.

One direction says a single coordinate can be no further off than the whole vector. The other says the whole vector can be no further off than all the coordinates put together.

Proof: vector limit $\Longrightarrow$ coordinate limits

Suppose $\lim_{t\to c} \vec{f}(t) = \vec{L}$; fix i. Choose $\epsilon > 0$. We want $\delta > 0$ with

$$0 < |t - c| < \delta \implies |f_i(t) - L_i| < \epsilon.$$

We can choose $\delta > 0$ so that $0 < |t - c| < \delta \implies \|\vec{f}(t) - \vec{L}\| < \epsilon$. But

$$\begin{aligned}\|\vec{f}(t) - \vec{L}\| &= \|(f_1(t) - L_1, \ldots, f_n(t) - L_n)\| = \sqrt{(f_1(t) - L_1)^2 + \cdots + (f_n(t) - L_n)^2} \\ &\geq \sqrt{(f_i(t) - L_i)^2} = |f_i(t) - L_i|,\end{aligned}$$

so the same δ gives $|f_i(t) - L_i| < \epsilon$. $\square$

Proof: coordinate limits $\Longrightarrow$ vector limit

On the other hand, assume $\lim_{t\to c} f_i(t) = L_i$ for each i; we want to show $\lim_{t\to c} \vec{f}(t) = \vec{L}$.

Choose $\epsilon > 0$. For each i we can find $\delta_i > 0$ such that

$$0 < |t - c| < \delta_i \implies |f_i(t) - L_i| < \frac{\epsilon}{n}.$$

Let $\delta = \min\{\delta_1, \ldots, \delta_n\}$, so that for $0 < |t - c| < \delta$ all n of these inequalities hold at once.

The idea: split $\vec{f}(t) - \vec{L}$ into n vectors, each with only one nonzero coordinate, and apply the triangle inequality

$$\|\vec{a} + \vec{b}\| \leq \|\vec{a}\| + \|\vec{b}\|.$$

Proof, continued

For $0 < |t - c| < \delta$,

$$\begin{aligned}
\|\vec{f}(t) - \vec{L}\| &= \|(f_1(t) - L_1, \ldots, f_n(t) - L_n)\| \\
&= \|(f_1(t) - L_1, 0, \ldots, 0) + \cdots + (0, \ldots, 0, f_n(t) - L_n)\| \\
&\leq \|(f_1(t) - L_1, 0, \ldots, 0)\| + \cdots + \|(0, \ldots, 0, f_n(t) - L_n)\| \\
&= |f_1(t) - L_1| + \cdots + |f_n(t) - L_n| \\
&< \frac{\epsilon}{n} + \cdots + \frac{\epsilon}{n} = \epsilon. \qquad \square
\end{aligned}$$

The example again, componentwise

$$\lim_{t\to 1}\left(\frac{t^2-1}{t-1},\ 2t+1\right) = (2,3),$$

one coordinate at a time:

$$\lim_{t\to 1}\frac{t^2-1}{t-1} = \lim_{t\to 1}(t+1) = 2, \qquad \lim_{t\to 1}(2t+1) = 3,$$

since $t+1$ and $2t+1$ are continuous (they are polynomials).

No ϵ, no δ — the proposition did that work once and for all. The hand computation was still worth doing once.

Derivatives

The derivative of a vector-valued function

Definition

If $\vec{f}\colon (a,b) \to \mathbb{R}^n$ and $c \in (a,b)$, define

$$\vec{f}\,'(c) = \lim_{h\to 0} \frac{\vec{f}(c+h) - \vec{f}(c)}{h} = \lim_{h\to 0} \frac{1}{h}(\vec{f}(c+h) - \vec{f}(c)),$$

when the limit exists.

The quotient makes sense: subtract two vectors, then multiply by the number $\frac{1}{h}$.

Meaning. Changing t from c to $c+h$ changes the value of the function by approximately $\vec{f}\,'(c)\,h$:

$$\vec{f}(c+h) \approx \vec{f}(c) + h\,\vec{f}\,'(c).$$

Derivatives are componentwise

Proposition

$\vec{f} = (f_1, \ldots, f_n)\colon (a, b) \to \mathbb{R}^n$ is differentiable at $t = c$ if and only if each f_i is differentiable at $t = c$, and then

$$\vec{f}\,'(c) = (f_1'(c), \ldots, f_n'(c)).$$

Proof idea.

$$\lim_{h\to 0} \frac{\vec{f}(c+h) - \vec{f}(c)}{h} = \lim_{h\to 0} \left(\frac{f_1(c+h) - f_1(c)}{h}, \ldots, \frac{f_n(c+h) - f_n(c)}{h} \right)$$
$$= \left(\lim_{h\to 0} \frac{f_1(c+h) - f_1(c)}{h}, \ldots \right) \quad \text{(componentwise limits).}$$

Example

$\vec{f}(t) = (2t,\ t^2 + 1)$, so

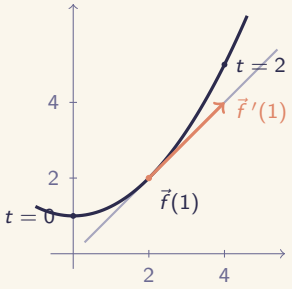
$$0 \mapsto (0,1), \quad 1 \mapsto (2,2), \quad 2 \mapsto (4,5).$$

Componentwise, $\vec{f}\,'(t) = (2,\ 2t)$.

At $t = 1$: $\vec{f}(1) = (2,2)$ and $\vec{f}\,'(1) = (2,2)$, so

$$\vec{f}(1+h) \approx (2,2) + h\,(2,2).$$

For instance
$\vec{f}(1.1) \approx (2,2) + (0.2, 0.2) = (2.2, 2.2)$; the true value is $(2.2, 2.21)$.



The same at $t = 2$

$\vec{f}\,'(2) = (2, 4)$ and $\vec{f}(2) = (4, 5)$, so

$$\begin{aligned}\vec{f}(2.3) &\approx \vec{f}(2) + \vec{f}\,'(2)\,(0.3)\\ &= (4, 5) + (2, 4)(0.3)\\ &= (4.6,\ 6.2).\end{aligned}$$

The true value is $\vec{f}(2.3) = (4.6,\ 6.29)$.

The arrow $\vec{f}\,'(c)$, drawn starting at the point $\vec{f}(c)$, points along the curve.

The rules

Proposition

Let $\vec{f}, \vec{g}\colon (a,b) \to \mathbb{R}^n$ be differentiable at $t = c$, and let $\lambda \in \mathbb{R}$. Then

$$\begin{aligned}(\vec{f}+\vec{g})'(c) &= \vec{f}\,'(c) + \vec{g}\,'(c),\\ (\lambda\vec{f})'(c) &= \lambda\,\vec{f}\,'(c),\\ (\vec{f}\cdot\vec{g})'(c) &= \vec{f}\,'(c)\cdot\vec{g}(c) + \vec{f}(c)\cdot\vec{g}\,'(c) \qquad \text{(Leibniz 1).}\end{aligned}$$

Here $\vec{f}\cdot\vec{g}\colon (a,b) \to \mathbb{R}$ is the real-valued function defined by

$$(\vec{f}\cdot\vec{g})(t) = \vec{f}(t)\cdot\vec{g}(t).$$

Leibniz 2

If $h\colon (a,b)\to\mathbb{R}$ and $\vec{f}\colon (a,b)\to\mathbb{R}^n$, define

$$h\vec{f}\colon (a,b)\to\mathbb{R}^n, \qquad (h\vec{f})(t) = h(t)\,\vec{f}(t).$$

If h and $\vec{f}$ are differentiable at c, then

$$(h\vec{f})'(c) = h'(c)\,\vec{f}(c) + h(c)\,\vec{f}\,'(c) \qquad \text{(Leibniz 2)}.$$

Proof of Leibniz 1

As before: $\vec{f}$ differentiable at $t = c \Rightarrow \vec{f}$ continuous at $t = c$ (componentwise, Lecture 10). Add and subtract $\vec{f}(c+h) \cdot \vec{g}(c)$ in the numerator:

$$\frac{\vec{f}(c+h) \cdot \vec{g}(c+h) - \vec{f}(c) \cdot \vec{g}(c)}{h}$$
$$= \vec{f}(c+h) \cdot \left(\frac{\vec{g}(c+h) - \vec{g}(c)}{h} \right) + \vec{g}(c) \cdot \left(\frac{\vec{f}(c+h) - \vec{f}(c)}{h} \right).$$

As $h \to 0$, $\vec{f}(c+h) \to \vec{f}(c)$ by continuity, and the quotients tend to $\vec{g}\,'(c)$ and $\vec{f}\,'(c)$. A dot product is a sum of products of coordinates, so the limit laws apply:

$$(\vec{f} \cdot \vec{g})'(c) = \vec{f}(c) \cdot \vec{g}\,'(c) + \vec{g}(c) \cdot \vec{f}\,'(c). \qquad \square$$

Notation

All of these mean the same thing:

$$\vec{f}\,'(t), \qquad \frac{d}{dt}\vec{f}(t), \qquad \frac{d\vec{f}}{dt};$$

and for a real-valued function,

$$f'(t), \qquad \frac{df}{dt}.$$

Example: curves on a sphere

For $\vec{f}\colon (a,b) \to \mathbb{R}^n$,

$$\begin{aligned}\frac{d}{dt}\|\vec{f}\|^2 &= \vec{f}\,'(t)\cdot\vec{f}(t) + \vec{f}(t)\cdot\vec{f}\,'(t)\\ &= 2\,\vec{f}(t)\cdot\vec{f}\,'(t),\end{aligned}$$

by Leibniz 1 applied to $\vec{f}\cdot\vec{f}$.

Now suppose the curve lies on a sphere of radius r centred at the origin, so $\|\vec{f}\|^2 = r^2$ is constant. Then its derivative is 0, so $\vec{f}\cdot\vec{f}\,' = 0$: the derivative $\vec{f}\,'(t)$ is perpendicular to $\vec{f}(t)$.

